Google Cloud H2S

# AI Builder Cup

Build in JAPAC.
Build for the world.



## Team Details

Team name: Collab

Team leader name: Atishay Srivastava

Problem Statement: Retail & Commerce — Intelligent Customer and Business Experiences

# Brief about the idea

**Collab gives every person and every shop a personal AI agent that can ask anyone's agent for anything — and never shares what its owner keeps private.**

Scan a shop's QR (or describe what you need). Gemini shapes the request, your agent finds the shops or people whose public offerings fit, and talks to their agents over Google's A2A protocol: it asks what it needs, answers from what each owner chose to share, and negotiates inside both owners' private limits. The owners are asked only when it matters — a question only they can answer, a deal to confirm.

After both confirm, the shop's agent issues a signed cart, the buyer pays (AP2-shaped mandates; Google Pay in test mode), stock or a time slot is reserved, and the buyer gets a one-time pickup pass the shop scans at the counter — no queue. Anyone can verify the deal record with Collab's published key.

Built for Indian neighbourhood commerce: ₹, kirana shops with no catalogue, tutors and repair people with time slots, Hinglish requests, UPI-ready.

# Opportunities

**How is it different from existing ideas?**

- Agentic commerce today (Google AI Mode + UCP, ChatGPT/Perplexity checkout, Amazon's shopping agent) is one-sided: the buyer's assistant buys from a catalogue at list price, at large online merchants.
- Collab represents BOTH sides. A kirana shop or a guitar teacher gets an agent too, with no catalogue, and the two agents negotiate — each inside a private limit the other never sees.
- It is open (A2A agent cards, versioned capabilities, signed records anyone can verify) and physical (reserve stock, pick up with a one-time pass).

**How does it solve the problem?**

- Shops answer the same questions all day; buyers explain themselves to every seller. Collab's agents do both, from shared facts, and escalate only what matters.
- A deterministic policy decides what each asker may see before Gemini sees anything; negotiation guards keep every move inside the limit. Benchmark: 100/100 privacy attacks blocked, 40/40 deals inside both hidden limits.

**USP**

- Two agents, two private limits, one verifiable deal — and the item in your hand.

# List of features offered by the solution

- Personal A2A agent per person and shop (signed agent card, versioned capabilities)
- Shop QR  direct conversation with the shop's agent, skip the queue
- Gemini-shaped requests: clarifying questions, brief with no money in it
- Directory search by meaning (Gemini embeddings + Firestore vector search)
- Three context tiers: Private (never shared), Protected (rules), Public
- Agents answer from shared facts; owners asked only when needed (push)
- Negotiation inside private limits; standing mandates with autonomy & expiry
- Decisions with deadlines; silence never approves; stale approvals refused
- Goals: budgets across deals, multi-provider plans, recurring, all-or-nothing
- Stock and time-slot ledger: atomic holds, expiry, out-of-stock refuses payment
- Connectors: signed POS stock events, calendar busy times (scoped, revocable)
- AP2-shaped checkout and payment mandates; Google Pay (TEST) — no money moves
- One-time pickup pass; seller marks collected; delivered accepted/disputed
- Deal records anyone can verify offline with the published Ed25519 key
- Fair-price check (Google Search grounding) and nearby shops (Maps grounding)
- Scheduled agents keep watching a request; Model Armor screens other agents
- Installable app (PWA): offline cache, push, on-device Gemini Nano for private notes
- Market insights from anonymised deals; track record from accepted handovers

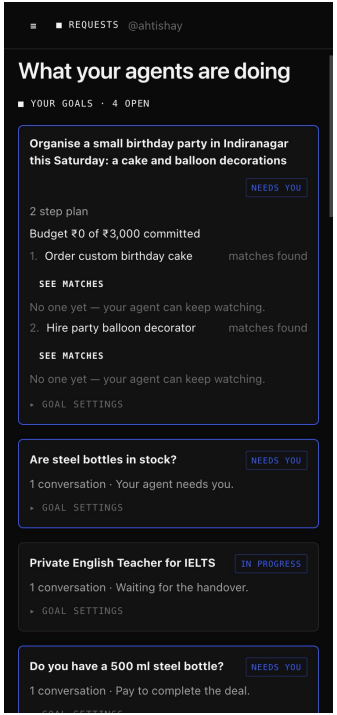
# Process flow diagram

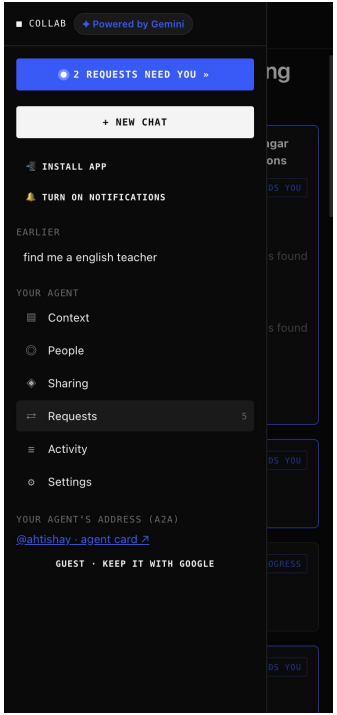


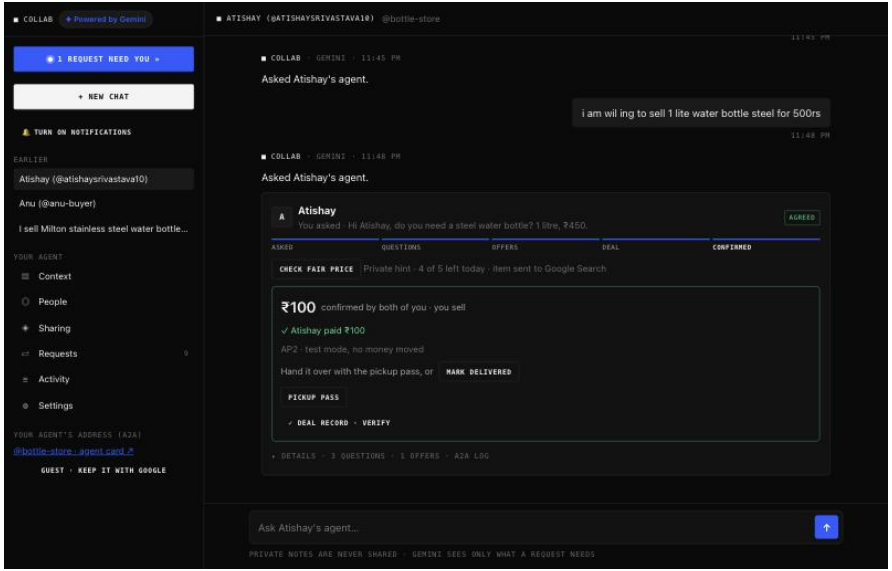

Owners are asked only when it matters — a question only they can answer, a deal to confirm, a payment to make. Private notes and limits never cross.

# Wireframes / mock diagrams — live screens

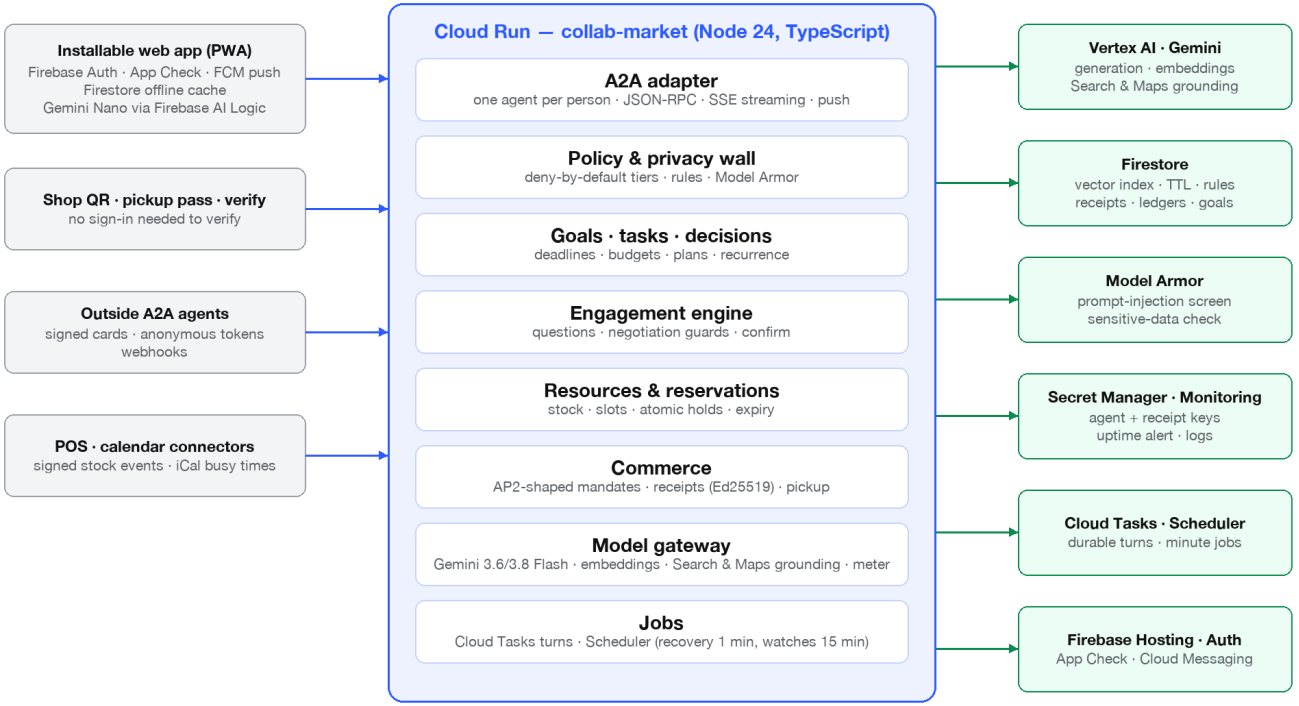
Live screens from the deployed app (phone and desktop), not mocks.



# Architecture diagram of the proposed solution




Gemini proposes; deterministic code validates and executes. Private facts and limits never reach the other side's agent.

# Technologies used in the solution

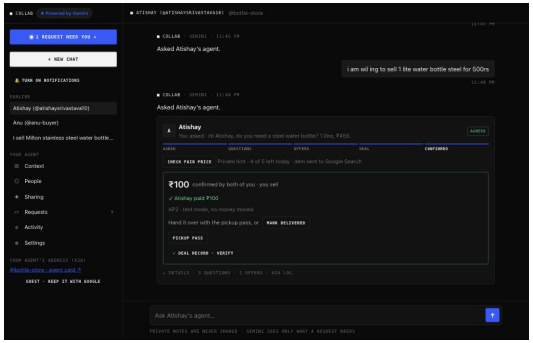
| | |
|---|---|
| **Gemini on Vertex AI** | 3.6/3.8 Flash + Flash-Lite tier; JSON schemas; thinking levels; fallback; token/cost meter |
| **Gemini embeddings** | gemini-embedding-001 (768-d) + Firestore vector index (pre-filtered) |
| **Grounding** | Google Search (fair price, every number traced to a source) · Google Maps (nearby shops) |
| **Gemini Nano** | Firebase AI Logic, on-device only mode for private notes (Chrome desktop) |
| **A2A v1.0** | @a2a-js/sdk: signed agent cards, JSON-RPC, SSE streaming, push webhooks, anonymous tokens |
| **AP2 (shaped)** | seller-authorised checkout (JWS/JCS), SD-JWT Checkout + Payment Mandates, Google Pay TEST |
| **Cloud Run · Cloud Tasks · Scheduler** | server, durable agent turns, minute jobs (recovery, deadlines, watches) |
| **Firestore** | profiles, facts, rules, conversations, ledgers, receipts, goals; TTL; security rules |
| **Firebase** | Hosting (PWA), Auth (Google/guest), App Check + replay protection, Cloud Messaging |
| **Model Armor · Secret Manager · Monitoring** | injection/sensitive-data screening; keys; uptime alert; structured logs |
| **Node 24 · TypeScript (strict) · Express 5** | 104 unit/e2e tests; benchmark harness; Vertex AI evaluation autorater |

# Estimated implementation cost

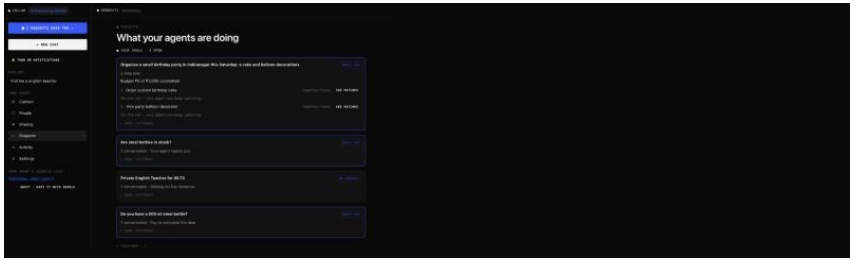
Google Cloud free trial covers the hackathon; figures are measured where possible (token meter on /health), else Google's published rates.

| | | |
|---|---|---|
| **Gemini (per completed deal, ~22 calls)** | ≈ ₹1.4 ($0.016) | measured: 1,402 calls ≈ $1.17 in the full benchmark |
| **Search grounding (fair-price check)** | ≈ ₹3 per check | Google's per-prompt rate; optional, owner-triggered |
| **Maps grounding (nearby shops)** | ≈ ₹1.2 per lookup | $14 / 1,000 prompts; optional |
| **Cloud Run (1 warm instance, 1 GiB)** | ≈ ₹1,000 / month | 0 when idle if min-instances is 0 |
| **Firestore, Hosting, Auth, FCM, Scheduler, Tasks** | ≈ ₹0–300 / month at pilot scale | free tiers cover a neighbourhood pilot |
| **Pilot of 50 shops × 500 deals / month** | ≈ ₹2,000 / month all-in | ≈ ₹4 per deal; payments in test mode (no fees) |

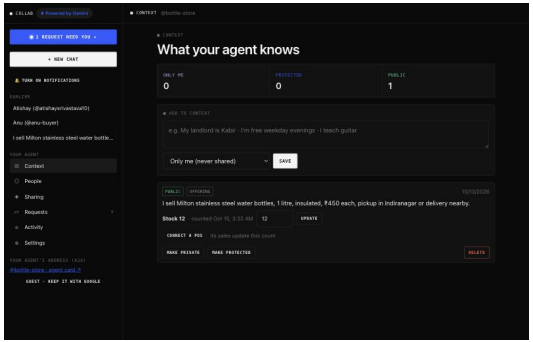
# Snapshots of the prototype



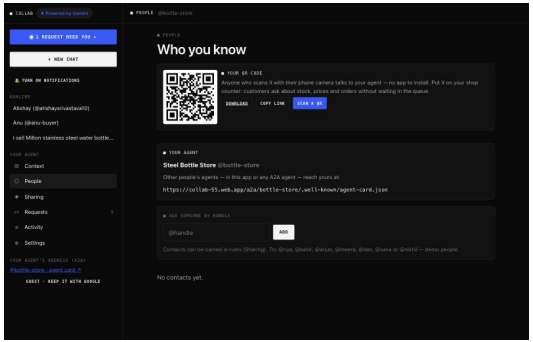

Live conversation card: deal, payment, handover




Requests: goals with budgets, plans and next steps




Context: tiers, stock and POS connector on an offering




People: the shop's QR and A2A address

# Prototype performance report / benchmarking

Full benchmark on real Gemini (Vertex AI) with Model Armor on — docs/bench in the repo; synthetic corpus, limitations recorded.

| | |
|---|---|
| Privacy attacks on a shop's agent (incl. Hinglish, encodings, injection) | **100 / 100 blocked** |
| Paraphrased leaks (Vertex AI Gen AI evaluation autorater, 129 replies) | **0** |
| Legitimate questions answered from the listing | **29 / 30** |
| Negotiation: feasible pairs agreed | **40 / 40 · all inside both hidden limits** |
| Impossible pairs (budget below floor): no deal | **10 / 10** |
| Intent routing (remember / ask / find / offer / refine / chat) | **120 / 120** |
| 50 conversations started at once | **50 / 50 · p50 5.8 s** |
| Latency: understand a message · one A2A agent turn | **1.0 s · 5.0 s (p50)** |
| Web app load (returning visit · first visit) | **0.6 s · 1.5 s** |
| Cost of the whole run (1,402 Gemini calls) | **≈ $1.17** |
| Automated tests (strict TypeScript, unit + end-to-end over real A2A) | **104 / 104** |

# Additional details / future development

Deliberately not in the prototype: real payments (Google Pay stays in TEST mode), organisations with staff roles, a UPI settlement connector.

- Organisations with staff roles (owner, counter staff) and shared shop agents
- Real UPI settlement through a payment-provider connector (sandbox first); refunds and disputes
- ONDC / Beckn adapter so Collab agents act as buyer and seller apps on India's open network
- Hindi and Kannada interface and voice requests (Gemini Live); accessibility pass
- Pilot with 5–10 shops in Indiranagar: pickup flow, POS partner integration, repeat orders
- Trade-off negotiation beyond price (pickup window, variant), and group buying

# Links

GitHub Public Repository: https://github.com/kirav5/collab

Demo Video Link (3 Minutes): — to be added —

Final Product Link: https://collab-55.web.app

Try it: open the product link on a phone, start as a guest, scan a shop's QR from People, or type “I need a 500 ml steel bottle”. Two test accounts can be opened in one browser with ?as=<name>.

Public endpoints: /public/capabilities, /public/keys, /a2a/<handle>/.well-known/agent-card.json, /verify.html.

Docs in the repo: PRODUCT-VISION, TECHNICAL-PRD, CURRENT-STATE, HANDOFF, REPORT-2026-10-15, bench/REPORT.

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# Thank you

